A risk-benefit analysis of French high fish consumption: a QALY approach.

The health risk and the nutritional benefit of a food are usually assessed separately. Toxicologists recommend limiting the consumption of certain fish because of methylmercury; while nutritionists recommend eating more oily fish because of omega 3. A common evaluation is imperative to provide coherent recommendations. In order to evaluate the risks along with the benefits related to fish consumption, a common metric based on the quality-adjusted life year (QALY) method has been used. The impact of a theoretical change from a medium n-3 PUFAs intake to a high intake is studied, in terms of the cardiovascular system (CHD mortality, stroke mortality and morbidity) and on fetal neuronal development (IQ loss or gain). This application can be considered as a sensitive analysis of the model used and looks at the impact of changing the dose-response relationships between cardiovascular diseases and n-3 PUFAs intakes. Results show that increasing fish consumption may have a beneficial impact on health. However, the confidence interval of the overall estimation has a negative lower bound, which means that this increase in fish consumption may have a negative impact due to MeHg contamination. Some limits of the QALY approach are identified. The first concerns determination of the dose-response relationships. The second concerns the economic origins of the approach and of individual preferences. Finally, since only one beneficial aspect and one risk element were studied, consideration should be given to how other beneficial and risk components may be integrated in the model.